# Assignment (Deadline Jesth 13)

## Short Question

1. What is computer graphics? Explain different application areas of computer graphics.
2. How can you draw a circle using mid-point circle algorithm? Explain with the algorithm.
3. Explain 3D basic geometric transformation with an example.
4. What is polygon clipping? Explain Sutherland Hodgman algorithm for polygon clipping.
5. Given a triangle with vertices A(2,3), B(5,5), C(4,3) by rotating 90 degree about the origin and then translating two unit in each direction. Use homogenous transformation matrix to find the new vertices of the triangle Note.
6. Define horizontal and vertical retrace of electron beam. Explain architecture of random scan display system.
7. How does decision parameter can be used to draw circle? Calculate the points to draw a circle having radius 5 and center (10,5) using midpoint circle algorithm.
8. What do you mean by window to viewport transformation? Explain 2D viewing pipeline.
9. Generate the points between the end points of a line (2,2) and (9,6) using DDA line drawing algorithm.
10. Apply Liang Barsky Line Clipping algorithm to the line with coordinates (30, 60) and (60, 25) against the window (xwmin, ywmin) = (10, 10) and (xwmax, ywmax) = (50, 50).
11. Use the Cohen-Sutherland algorithm to clip the line P1(70, 20) and P2(100, 10) against a window lower left hand corner (50, 10) and upper right hand corner (80, 40).
12. A homogenous coordinate point P(3, 2, 1) is translated in x, y, z direction by -2, -2 & -2 unit respectively followed by successive rotation of 600 about x- axis. Find the final position of homogenous coordinate.

## Long Question

1. Explain the working details of DDA line drawing algorithm? Digitize the line with endpoints (2,2) and (10,5) using Bresenham's line drawing algorithm.

2. Derive the formula for windows to viewport transformation. Given a window bordered by (0,0) at the lower-left and (4,4) at the upper right. Similarly, a viewport bordered by (0,0) at the lower-left and (2,2) at the upper right. If a window at position (2,4) is mapped into the viewport.
3. What will be the position of viewport to maintain same relative placement as in window?
4. What is the need of homogeneous coordinate system in geometric transformation system? Find the new coordinates of rectangle ABCD whose center is at (4,2), reduced to half of its size while the center remains same. Coordinates: A(0,0), B(0,4), C(8,4), D(8,0).
5. Explain Sutherland Hodgeman Polygon Clipping algorithm and trace it to clip a polygon with end points P(10,30), Q(25,35), R(35,50), S(60,30), T(45,28), U(40,15) against a window whose lower-left corner is (15,25) and upper-right corner is (50,40).